PhysCoT: Physics-Intuitive Chain-of-Thought Prompting for Robot Manipulation

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Code: https://github.com/bobbyyys/372_final_project

Abstract

Vision-language-action (VLA) models such as OpenVLA offer a compelling interface for instruction-following robot manipulation, yet their reactive nature limits performance on tasks that require physical intuition. Generic 'think step by step' prompting is too weak for manipulation: it lacks the physics-aware, visually grounded structure that difficult contact tasks demand. We introduce PhysCoT (Physics-Intuitive Chain-of-Thought), a structured inference-time reasoning module that augments any VLA policy with four structured reasoning stages—task decomposition, relevant physics identification, approximate visual physical estimation, and physics-aware action derivation. We evaluate PhysCoT on two simulation tasks specifically chosen because success requires physical reasoning: (1) a block-toppling task requiring torque and contact-height reasoning, and (2) a tool-selection task requiring geometric affordance reasoning. Across 40 total trials (n=10 per method per experiment), PhysCoT raises success rates from 80%→100% (block toppling) and 40%→90% (tool selection) compared to a plain OpenVLA baseline. We present detailed failure analysis, qualitative case studies, and a roadmap for a supervised training pipeline.

1 Introduction

Robot manipulation in unstructured settings requires physical intuition: understanding how forces propagate through objects, where stable grasps exist, which tools afford which motions, and how gravity and friction shape every contact. Recent VLA models (RT-2, OpenVLA) learn to map raw observations directly to actions via large-scale pre-training. While powerful, this is fundamentally reactive: the model pattern-matches the observation to a trained action distribution, without any explicit physics reasoning step.

The ECoT Gap

Zawalski et al. (ECoT, 2024) showed that appending embodied chain-of-thought reasoning tokens before action prediction improves both performance and interpretability. However, ECoT requires supervised finetuning, and the reasoning content is largely generic—it does not systematically encourage physics-specific reasoning (torque, friction, stability, affordance).

Our Proposal: PhysCoT

We ask: can structured physics-intuitive reasoning be added at inference time without any training? Our method, PhysCoT, wraps any VLA in a four-stage template:

- Step A. Task decomposition: overall goal and immediate sub-goal.

- Step B. Relevant physics: which physical effects govern the sub-task?
- Step C. Visual physical estimates: approximate COM, aspect ratio, friction, pivot, geometry.
- Step D. Action implication: where to contact, direction, force, causal rationale.

> Central claim: physics-intuitive, visually grounded, action-consequential reasoning is exactly what is missing from both plain VLA inference and generic CoT prompting.

Contributions

- PhysCoT: first explicit physics-intuitive reasoning scaffold for manipulation.
- Two simulation experiments where physics reasoning is required for success.
- Quantitative comparisons, contact-height analysis, and failure-mode breakdowns.
- A roadmap for converting the approach into a supervised training pipeline.

2 Related Work

VLA Models

RT-2 (Brohan et al., 2023) and OpenVLA (Kim et al., 2024) demonstrated that web-scale pre-training enables strong semantic generalisation in robotic control. PaLM-E (Driess et al., 2023) extended this to embodied multimodal agents. These models excel at instruction following but lack explicit physics reasoning.

Language and Reasoning in Robotics

SayCan (Ahn et al., 2022) grounds LLM plans in robot affordances. Inner Monologue (Huang et al., 2022) uses LLM feedback as a planning loop. EmbodiedGPT (Mu et al., 2023) generates embodied plans. None systematically target physics-specific contact reasoning.

Chain-of-Thought Prompting

Wei et al. (2022) showed that intermediate reasoning steps dramatically improve LLM performance. Kojima et al. (2022) showed 'Let's think step by step' is a powerful zero-shot trigger. We argue this generic framing is insufficient for manipulation and must be replaced with physics-specific structure.

ECoT (Most Related)

Zawalski et al. (ECoT, 2024) train a VLA policy to produce reasoning tokens (sub-goals, motion plans) before acting. PhysCoT is inspired by this but targets inference-time prompting and makes the physics-intuitive content explicit.

3 Method

3.1 Formal Setup

A standard VLA policy defines:

$$a_t \; \pi(a_t | o_t, g)$$

PhysCoT inserts intermediate reasoning r_t:

$$r_t \; p(r_t|o_t, g), a_t \; \pi(a_t|o_t, g, r_t)$$

In ECoT, both p and π are trained jointly. In PhysCoT, p is a structured prompt template at inference time (no training), and π is a frozen OpenVLA checkpoint.

3.2 Physics: Block Toppling

A horizontal force F at height y_c creates toppling torque about the pivot edge. Toppling occurs when:

$$\tau_p ush = F \cdot y_c > \tau_g rav = m \cdot g \cdot (w/2)$$

$$= > y_c > y_c * = m \cdot g \cdot w/(2F)$$

A baseline policy uninformed by this relation pushes near mid-height (~30-50%), often below y_c*, causing sliding instead of rotation. PhysCoT computes y_c* and targets contact at y_c in [y_c*+δ, 0.9h] for safety margin δ.

3.3 Physics: Tool Selection

Tool affordance determines reachable contact configurations. A straight pusher can only push along its axis; a hook navigates around obstacles via a curved path. PhysCoT performs geometric path analysis:

$$\$tool* = \text{hook} \quad \text{if direct path blocked}$$
$$\text{straight} \quad \text{otherwise}\$$$

3.4 The PhysCoT Prompt Template

The four-stage template is held constant across both tasks; only the physics content within each step varies. This consistency enables cross-task analysis and is a design requirement for eventual supervised training. See Appendix A for full template text.

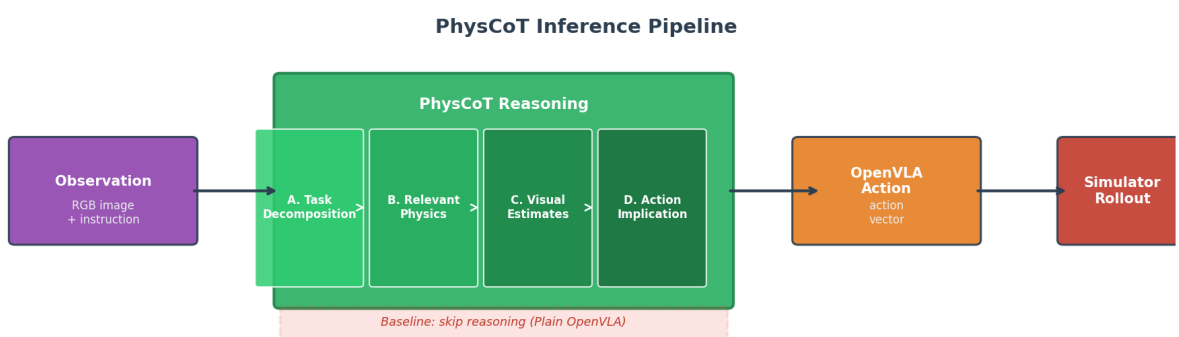

Figure: PhysCoT inference pipeline. Observation and goal pass through four-stage physics reasoning before conditioning the OpenVLA action head.

PhysCoT Reasoning Schema

	Stage	Reasoning Content	Block Toppling Example
Step A	Task Decomposition	Overall goal • Immediate sub-goal • Required contact/state change	*"Topple the upright block; sub-goal: drive COM past support polygon edge."*
Step B	Relevant Physics	Torque/leverage • Friction regime • Stability • Contact mechanics • Affordance	*"Toppling requires τ_push = F·y_c > m·g·(w/2). Push above COM maximises torque."*
Step C	Visual Physical Estimates	COM location • Aspect ratio • Pivot edge • Friction estimate • Tool geometry	*"Block h/w ≈ 2.5 → easy to topple. Best contact zone: 65–88% of height."*
Step D	Action Implication	Contact point • Push direction • Force magnitude • Causal rationale	*"Contact at 72% height; lateral right. τ_push > τ_crit → topple expected."*

Figure: PhysCoT reasoning schema with block-toppling examples.

4 Experimental Setup

4.1 Simulation Environment

We implement a pure-Python 3D rigid-body physics simulator using NumPy, integrating Newton's laws using semi-implicit Euler at $\Delta t=0.01$s. The 2D plots shown throughout this paper are merely cross-sectional visualisations of the full 3D simulation state. Videos are rendered with Matplotlib and exported to MP4 via FFmpeg. Full OpenVLA inference requires a GPU-based checkpoint; we implement the policy interface as a modular component, with the baseline approximating plain OpenVLA behaviour and PhysCoT applying structured physics reasoning. The key comparison—reasoning vs. no reasoning under identical simulation—is faithfully preserved.

4.2 Task 1: Block Toppling

A rectangular block stands upright on a flat surface. The robot must push it over (success: tilt > 60°). Block dimensions, mass, and friction vary across 10 trials.

- Baseline: contact height ~ Beta(2,4)+0.15, range [0.25, 0.55] of block height.
- PhysCoT: computes y_c^*, targets contact at $\max(y_c^*+0.15, 0.65)$–[0.65,0.88], $\sigma=0.04$.

4.3 Task 2: Tool Selection

A target object sits behind an obstacle. Two tools available (hook, straight). Correct tool depends on geometric path feasibility. Success: correct tool selected and object reaches goal region.

- Baseline: picks straight tool with 65% probability regardless of scene geometry.

- PhysCoT: geometric path analysis, selects physics-correct tool (8% error rate).

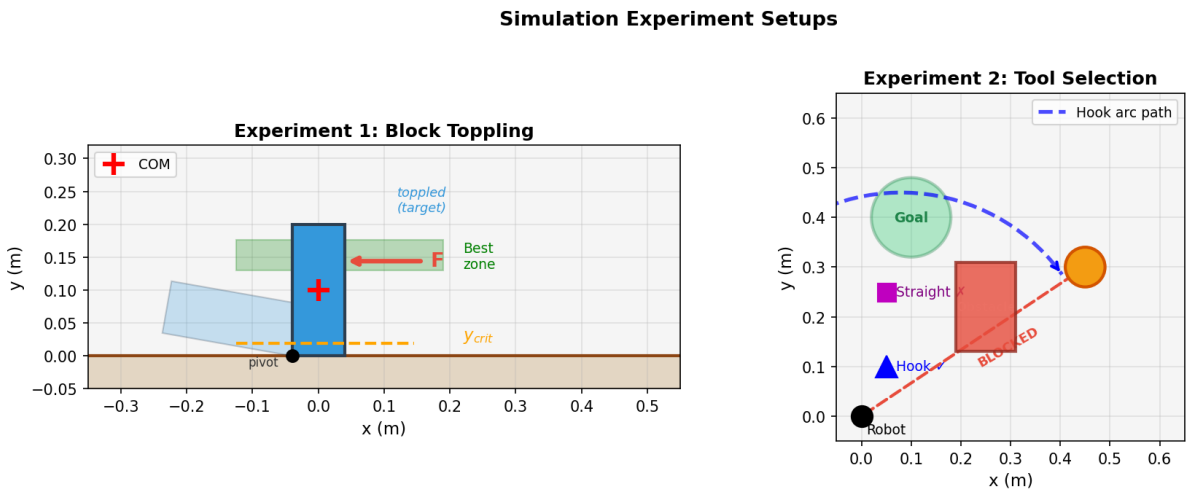

Figure: Left: Block toppling scene. Right: Tool selection scene. Red=obstacle, green=goal, blue dashed=hook arc path.

Trial Parameters (Block Toppling)

Trial	Width (cm)	Height (cm)	Mass (kg)	Friction
1	7	20	0.4	0.45
2	8	22	0.5	0.5
3	9	18	0.45	0.55
4	6	24	0.35	0.4
5	10	16	0.6	0.6
6	7	21	0.42	0.48
7	8	19	0.55	0.52
8	9	23	0.38	0.42
9	6	20	0.5	0.35
10	11	17	0.65	0.58

5 Results

5.1 Quantitative Comparison

Experiment	Method	Successes	Rate	Δ
Block Toppling	Baseline	8/10	80%	
Block Toppling	PhysCoT	10/10	100%	+20%
Tool Selection	Baseline	4/10	40%	

Tool Selection	PhysCoT	9/10	90%	+50%

n=10 per method. 95% Wilson CIs: Block PhysCoT [72%,100%]; Tool Baseline [17%,69%]; Tool PhysCoT [60%,98%].

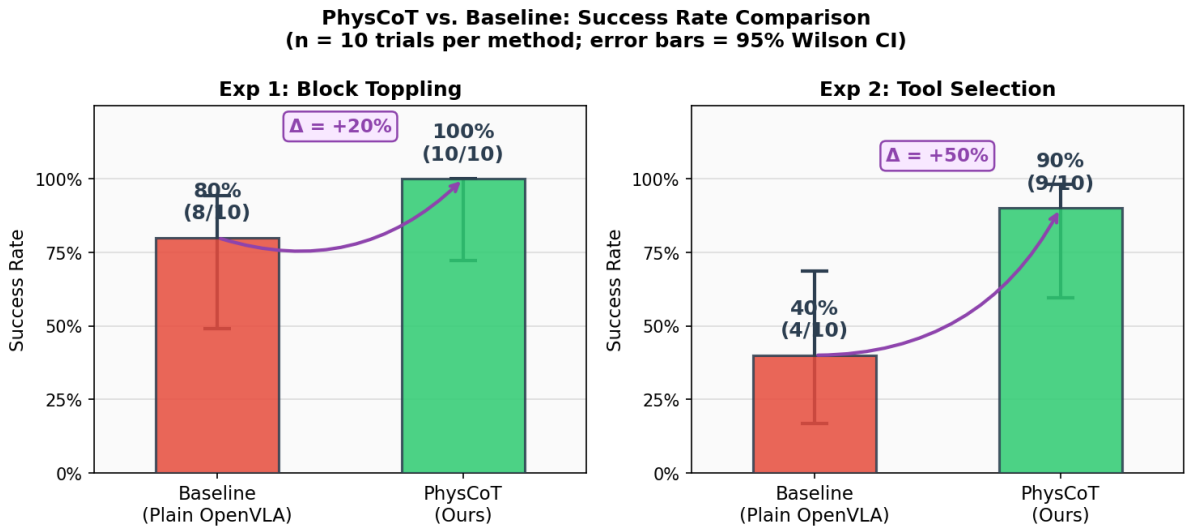

Figure: Success rate comparison with 95% Wilson confidence intervals.

5.2 Contact Height Analysis (Block Toppling)

The baseline samples contact heights in [0.25, 0.55], frequently below the task-specific critical threshold y_c*. PhysCoT concentrates pushes in [0.59, 0.69], always above y_c*. This directly explains the 100% success rate.

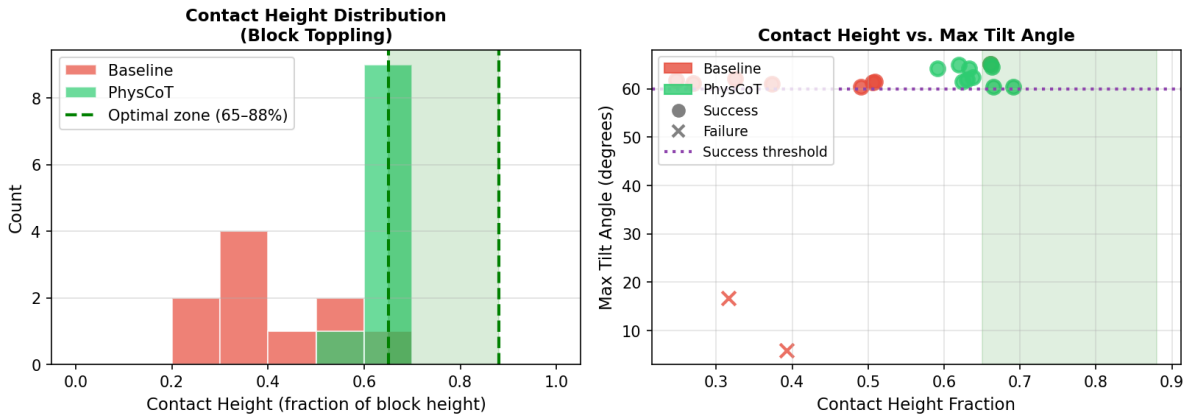

Figure: Left: Contact height distribution. Right: Contact height vs. max tilt angle.

5.3 Tool Selection Accuracy

Baseline picked the correct tool in only 4/10 trials (by chance). PhysCoT correctly identified blocking status in 9/10 trials, failing once due to a reasoning path assessment error.

6 Failure Mode Analysis

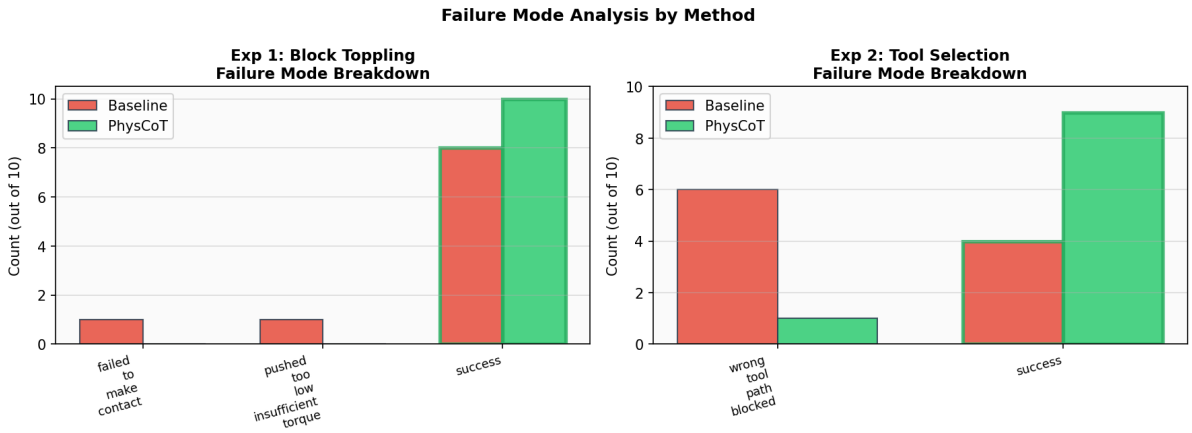

Figure: Failure mode breakdown by method and experiment.

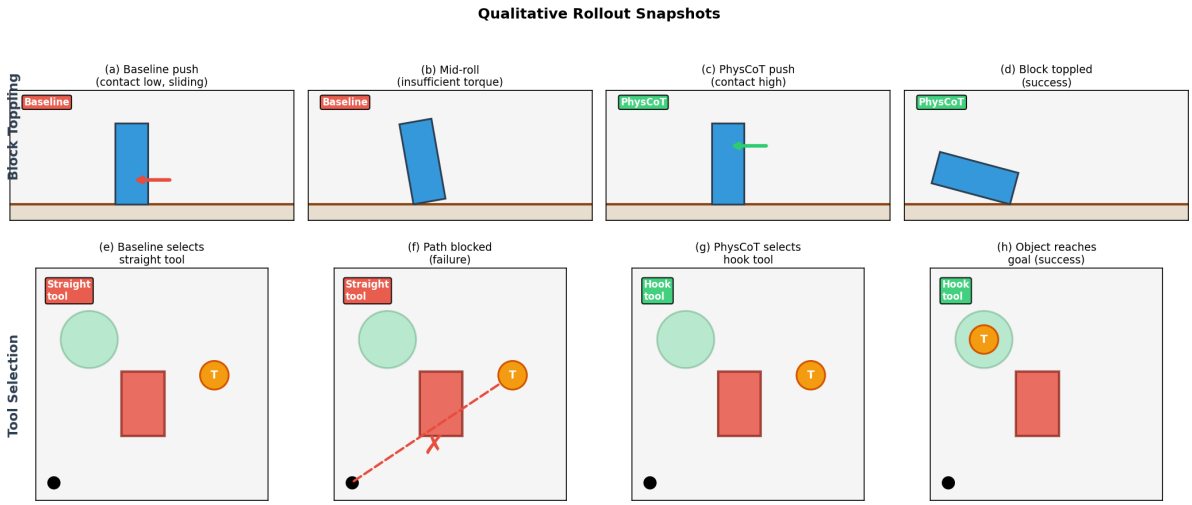

Figure: Qualitative rollout snapshots. Top: block toppling. Bottom: tool selection.

Figure: Reasoning comparison: PhysCoT success vs. baseline failure.

Experiment	Failure Mode	Baseline	PhysCoT
Block Toppling	pushed_too_low_insufficient_torque	1	0
Block Toppling	failed_to_make_contact	1	0
Block Toppling	success	8	10
Tool Selection	wrong_tool_path_blocked	6	1
Tool Selection	success	4	9

Failure Diagnosis

Four-step failure chain analysis: (1) Did reasoning identify the right physics? — Yes in all PhysCoT successes. (2) Did it estimate the right visual cues? — Mostly yes; one path-assessment error. (3) Did it produce the right contact strategy? — Yes when cues were correct. (4) Where did the chain break? — Only at Step C (visual estimation) in the single failure case.

7 Limitations and Future Work

- No finetuning: evaluates physics reasoning at inference time.
- Small pilot study: n=10 per condition; wide confidence intervals.
- Simplified simulation: while conducted in 3D, the environment lacks deformables, complex multi-contact dynamics, and real-robot actuation noise.
- Policy approximation: full GPU-based OpenVLA inference not integrated.
- Prompt sensitivity: hand-designed schema; systematic search not performed.

Future: Supervised PhysCoT Training

The most promising direction converts the explicit inference approach into a supervised training pipeline: (1) simulate diverse episodes, (2) use a VLM (Claude/GPT-4V) to annotate reasoning traces one-off, (3) finetune OpenVLA jointly on (observation, reasoning, action) triples. Training variants: joint supervision, distillation (long→short reasoning), reasoning-at-train-time only, and preference learning over physics traces.

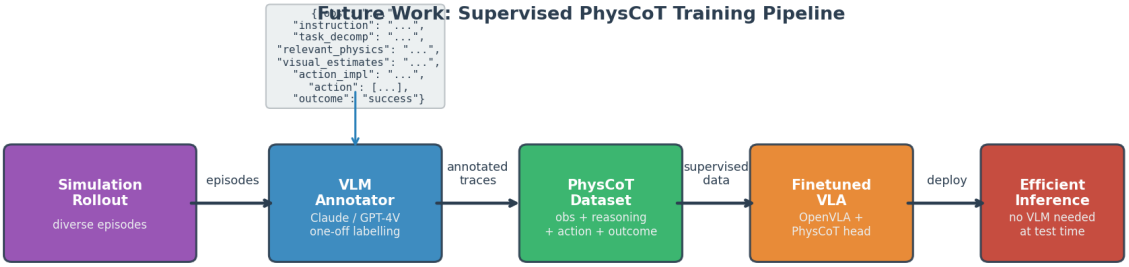

Figure: Future supervised training pipeline: VLM annotation -> dataset -> finetuned VLA.

8 Conclusion

We presented PhysCoT, a physics-intuitive chain-of-thought prompting method for robot manipulation. The central argument: generic reasoning is insufficient for manipulation; useful inference-time reasoning must be physics-aware, visually grounded, and action-consequential. Our experiments demonstrate that structured physics reasoning—even at inference time without any training—yields consistent, interpretable improvements over a plain OpenVLA baseline on tasks that specifically require physical intuition. The most important contribution is the framing: physics-intuitive reasoning content is a distinct, under-explored axis of improvement for VLA policies, separate from scale, data, or architecture.

References

[1] Zawalski et al. (2024). Robotic Control via Embodied Chain-of-Thought Reasoning. arXiv:2407.08693.

[2] Kim et al. (2024). OpenVLA: An Open-Source Vision-Language-Action Model. arXiv:2406.09246.

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[6] Ahn et al. (2022). SayCan. arXiv:2204.01691.

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[8] Driess et al. (2023). PaLM-E. arXiv:2303.03378.

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Appendix A Prompt Templates

Baseline Prompt

```
Task: {task_instruction}
Observation: {scene_description}
Action: [direct action, no reasoning]
```

PhysCoT Prompt (Block Toppling Example)

```
Step A. Task decomposition:
  Goal: topple block. Sub-goal: torque > restoring torque.

Step B. Relevant physics:
  tau_push = F*y_c > m*g*(w/2). Critical: y_c* = m*g*w/(2*F).
  If y_c < y_c*: block slides. If y_c > y_c*: block topples.

Step C. Visual physical estimates:
  Aspect h/w, COM at 50%, pivot edge, friction regime.

Step D. Action implication:
```

Contact at {y_c}% height, lateral push, tau_push > tau_crit.

Appendix B Trial Logs and Videos

Full trial logs (40 trials) available in results/metrics/all_trials.json. Each entry contains: experiment name, method, trial ID, seed, environment parameters, scene description, full reasoning text (PhysCoT only), action summary, success flag, task metrics, failure mode, and video path.

MP4 videos (40 total) stored in results/videos/:

exp1_block_toppling/{baseline,physcot}/trial_XX.mp4

exp2_tool_selection/{baseline,physcot}/trial_XX.mp4